

README

This document describes the material stored in the OSF repository <https://osf.io/g96f4/>, and described in detail here:

Chierchia, G., Fuhrmann, D., Knoll, L. J., Pi-Sunyer, B. P., Sakhardande, A. L., & Blakemore, S. J. (2019). The matrix reasoning item bank (MaRs-IB): novel, open-access abstract reasoning items for adolescents and adults. *Royal Society Open Science*, 6(10), 190232. DOI: <https://doi.org/10.1098/rsos.190232>. *joint 1st authors

In addition to this “README” document, and the manuscript, the repository consists of two main branches: the “Items” branch provides MaRs-IB items, and the “Item-level norms” branch provides item-level breakdown of performance data. These two branches are described in more detail below.

Reuse

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Please note that our task is not an IQ test. It is not intended to be used to determine someone’s intelligence or cognitive ability in, for example, educational, clinical or commercial contexts. This is because we do not have population norms for our task, so it does not generate an IQ score.

Items

Three item sets are available on the OSF repository. Item sets are identical apart from the item colors: “Item set 1” is described in the study referenced above (Chierchia et al., 2019). All item-level statistics pertain to this item set. “Item set 2” and “Item set 3” use two different sets of color-blind friendly colors and have not been tested. Color palettes for these sets were drawn from (Wong et al., 2011).

Each item set consists of three test forms (called “test form 1”, “test form 2” and “test form 3”) (see Figure 1). The latter are also identical and vary only with the respect to the shapes used (see Chierchia et al., 2019 for details). Item labels begin with one of the following: “tf1”, “tf2” or “tf3”, which indicates the test form they pertain to. Item labels also contain one of the following: “ss1”, “ss2” or “ss3”, which indicates the shape set they pertain to.

Each test form comprises 80 items (i.e., puzzles). Each individual item comprises of nine images – one 3x3 matrix and eight possible solutions (yielding 720 images in “jpeg” format). Matrix images can also be identified by their names, which always contain the following string: “_M_” (for “matrix”). The possible solutions can also be identified by their name, which always contain one of the following: “_T1_”, “_T2_”, “_T3_” and “_T4_”. The solution containing the string “_T1_” identifies the correct solution. Finally, the four solutions of each puzzle are available in two versions depending on the distractor algorithm used to generate them: half will contain the string “_md_”, as in “minimal difference” and the remaining half will contain the string “_pd_”, as in “paired difference” (see Chierchia, Fuhrmann et al., 2018 for details on the distractor strategies).

Each of the 80 items has a dimensionality score (provided in the table “Item dimensionality.csv”), which predicts its difficulty (see manuscript). The association between items and dimensionality score is fixed across test forms, shape sets and distractor strategy.

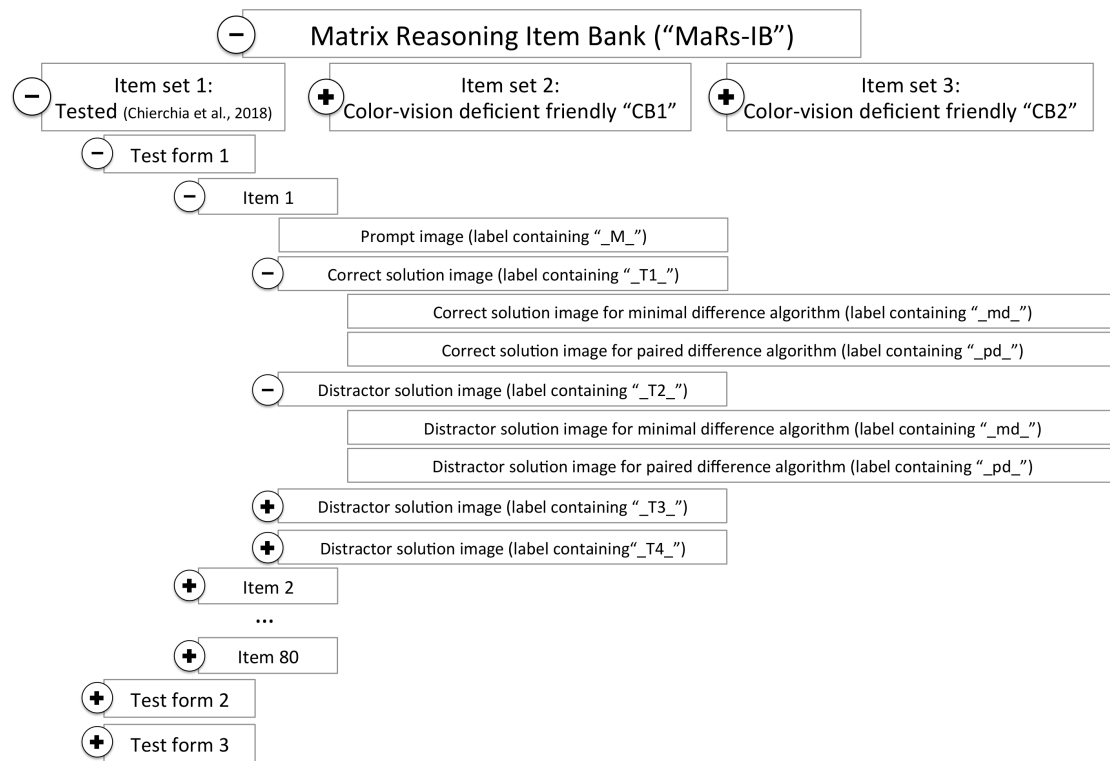


Figure 1. Repository Structure. All closed branches (“+” symbols) contain the same structure of elements as the open ones (“-” symbols).

Item-level statistics

As described in the manuscript, the OSF repository provides three tables containing item-level performance norms. Each table contains a set of performance-related statistics for each of the 80 items. One table breaks down the statistics by test form (“[item-level statistics by test form.csv](#)”), the second by shape set (“item-level statistics by shape set.csv”) and the third by age group (“item-level statistics by age group.csv”). These item-level statistics can help researchers to customize the relational reasoning task to their needs (e.g., by juxtaposing easier vs. harder items, while perhaps adapting this to specific age groups). Table 1 below shows the first few rows of the item-level statistics by shape set as an example. For example, the row containing the number 2 in the “Position” column and containing “ss1” in the “Group” column, corresponds to the stimuli labels beginning with “2” and containing the string “_ss1_”. In the table “item-level statistics by age group”, age groups are labelled using the abbreviations used in the manuscript: “YA” = younger adolescents, “MA” = mid adolescents, “OA” = older adolescents and “Ad” = adults.

Group	Item	N	Percentage .correct	SE.correct	RT.median.corr	RT.IQR.corr	IES
total	1	648	97	1	2412	1579	2489
ss1	1	226	98	1	2398	1537	2452
ss2	1	212	97	1	2432	1660	2502
ss3	1	210	96	1	2406	1525	2514
total	2	654	97	1	4447	2513	4566
ss1	2	215	98	1	3933	2261	4008
ss2	2	228	95	1	4902	2504	5150
ss3	2	211	99	1	4458	2595	4501

Table 1. Structure of the Item-Level Statistics of the MaRs-IB. The table exemplarily shows the first two items of the table “item-level statistics by shape set”. The other tables in the repository are structured identically. *Column names:* “Item” identifies the items provided in the item sets. For example, rows containing the number “2” in the Item column, correspond to repository item labels beginning with the number 2; “N” indicates the number of participants that completed the given item; Percentage.correct = percentage of correct responses (accuracy); SE.correct = standard error of response accuracy; “RT.median.corr” = median response time on correct trials; “RT.IQR.cor” = inter-quartile range of median response times on correct trials; “IES” = inverse efficiency.

References

Chierchia, G., Fuhrmann, D., Knoll, L. J., Pi-Sunyer, B. P., Sakhardande, A. L., & Blakemore, S. J. (2019). The matrix reasoning item bank (MaRs-IB): novel, open-access abstract reasoning items for adolescents and adults. *Royal Society Open Science*, 6(10), 190232. DOI: <https://doi.org/10.1098/rsos.190232>. *joint 1st authors

Wong, B. (2011). Color blindness. *Nature Methods*, 8(6), 441.